

Lab Sheet 07

Load the Data

First, we will load the 'Hitters' dataset from the ISLR library and handle any missing values.

```
library(ISLR)
data("Hitters")
sum(is.na(Hitters)) # Check for missing values
```

```
Hitters = na.omit(Hitters) # Remove rows with missing values
nrow(Hitters) # Number of rows after cleaning
```

Best Subset Selection

We will use the leaps package to perform Best Subset Selection and evaluate the model performance using different metrics such as RSS, R-squared, Cp, BIC, and Adjusted R-squared.

```
library(leaps)
best = regsubsets(Salary~., data = Hitters, nvmax = 19) # Perform Best Subset Selection
results = summary(best)

# Model selection metrics
RSS = results$rss
r2 = results$rsq
Cp = results$cp
```

```
BIC = results$bic
```

```
Adj_r2 = results$adjr2
```

```
cbind(RSS, r2, Cp, BIC, Adj_r2) # Combine and display metrics
```

```
# Identify the best models
```

```
which.min(Cp) # Model with minimum Cp
```

```
which.min(BIC) # Model with minimum BIC
```

```
which.max(Adj_r2) # Model with maximum Adjusted R-squared
```

```
coef(best, 10) # Coefficients for the model with 10 predictors (based on Cp)
```

```
coef(best, 6) # Coefficients for the model with 6 predictors (based on BIC)
```

```
coef(best, 11) # Coefficients for the model with 11 predictors (based on  
Adjusted R-squared)
```

Forward Stepwise Selection

Next, we will use the Forward Stepwise Selection method to select the best subset of predictors.

```
fwd = regsubsets(Salary~., data = Hitters, nvmax = 19, method = "forward")
```

```
results = summary(fwd)
```

```
# Model selection metrics
```

```
RSS = results$rss
```

```
r2 = results$rsq
```

```
Cp = results$cp
```

```
BIC = results$bic
```

```
Adj_r2 = results$adjr2
```

```
cbind(RSS, r2, Cp, BIC, Adj_r2)
```

```
# Identify the best models
```

```
which.min(Cp)
```

```
which.min(BIC)
```

```
which.max(Adj_r2)
```

```
coef(fwd, 10) # Coefficients for the model with 10 predictors (based on Cp)
```

```
coef(fwd, 6) # Coefficients for the model with 6 predictors (based on BIC)
```

```
coef(fwd, 11) # Coefficients for the model with 11 predictors (based on  
Adjusted R-squared)
```

Backward Stepwise Selection

Finally, we will use the Backward Stepwise Selection method to identify the best subset of predictors.

```
bwd = regsubsets(Salary~., data = Hitters, nvmax = 19, method = "backward")
```

```
results = summary(bwd)
```

```
# Model selection metrics
```

```
RSS = results$rss
```

```
r2 = results$rsq
```

```
Cp = results$cp
```

```
BIC = results$bic
```

```
Adj_r2 = results$adjr2
```

```
cbind(RSS, r2, Cp, BIC, Adj_r2)
```

```
# Identify the best models
```

```
which.min(Cp)
```

```
which.min(BIC)
```

```
which.max(Adj_r2)
```

```
coef(bwd, 10) # Coefficients for the model with 10 predictors (based on Cp)
```

```
coef(bwd, 8) # Coefficients for the model with 8 predictors (based on BIC)
```

```
coef(bwd, 11) # Coefficients for the model with 11 predictors (based on  
Adjusted R-squared)
```